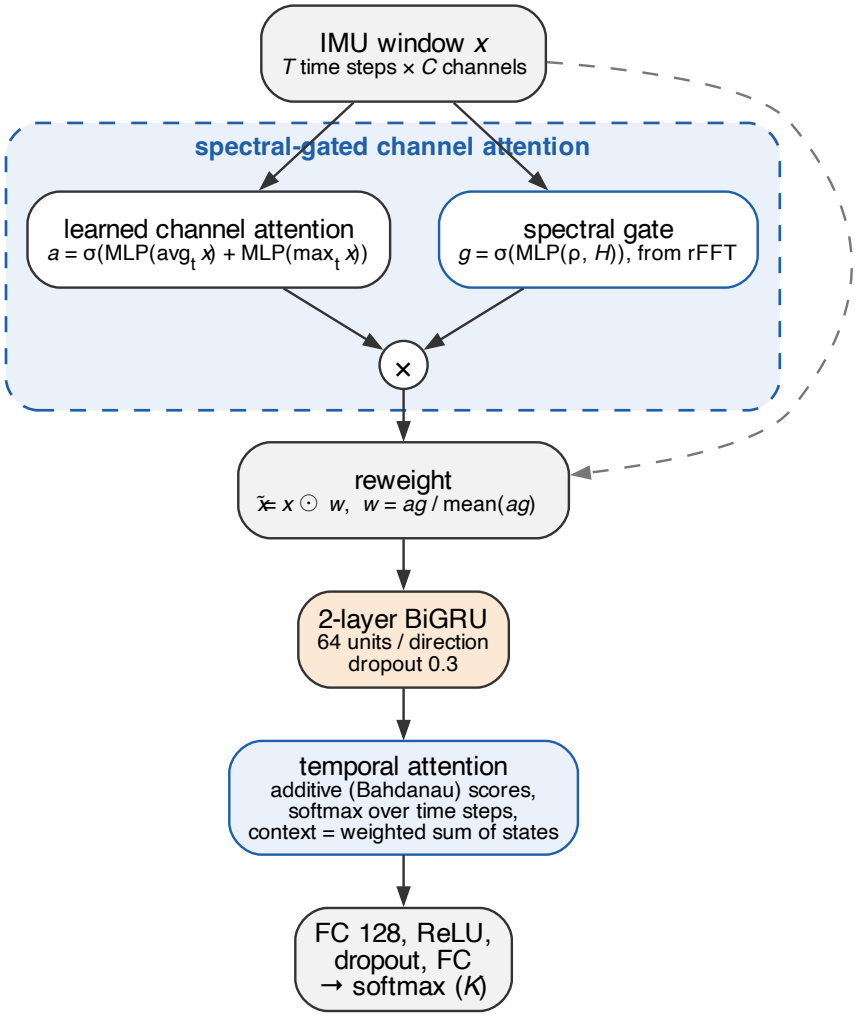

IMU window x
T time steps × C channels
spectral-gated channel attention
learned channel attention
a = σ(MLP(avg_t x) + MLP(max_t x))
spectral gate
g = σ(MLP(ρ, H)), from rFFT
×
reweight
x̃ = x ⊙ w, w = ag / mean(ag)
2-layer BiGRU
64 units / direction
dropout 0.3
temporal attention
additive (Bahdanau) scores,
softmax over time steps,
context = weighted sum of states
FC 128, ReLU,
dropout, FC
→ softmax (K)